Policy Title: Procedural Sedation, (Moderate & Deep), by Non-Anesthesiology Personnel	**Policy Number:** SCR.QRM.PCS.016
Regional Business Owner Title: S V P COO Clinical Operations	
Regional Custodian Title: Regional Director, Accreditation Regulation & Licensing	**Original Effective Date:** June 2003
Medical Center Business Owner Title: Anesthesia Chief / Chief Nurse Executive	
Medical Center Custodian Title: Medical Center Administration Team (MCAT)	**Revision Effective Date:** November 6, 2015
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1.0 Policy Statement

It is the policy of Kaiser Permanente to maintain patient safety during the administration of moderate and deep sedating medications.

2.0 Purpose

This document establishes policy for Kaiser Permanente's Southern California Region for procedural sedation, (moderate and deep) by Non-Anesthesiology Personnel.

3.0 Scope/Coverage

3.1 This policy applies to all employees who are employed by any of the following entities (collectively referred to as "Kaiser Permanente"):

3.1.1 Kaiser Foundation Hospitals;

3.1.2 Southern California Permanente Medical Group (SCPMG)

3.2 Region-wide Standard of Care

3.2.1 This policy applies to procedural sedation, (moderate and deep) practices in all settings (Kaiser Foundation Hospitals and SCPMG Medical Offices) in the Southern California Region.

3.3 Covered Clinical Departments

3.3.1 This policy addresses all departments that administer moderate and deep sedation across the Medical Center Service Area. Activities of the Departments of Anesthesiology are governed by their own policies, procedure and privileges, which establish equivalent standards of care.

3.4 Medical Center Policies and Procedures

3.4.1 Each medical center shall establish and maintain policies and procedures that meet the standards of care established by this policy.

4.0 Definitions

4.1 Levels of Sedation:

4.1.1 **Minimal Sedation (Anxiolysis)** - Minimal Sedation is a drug-induced state during which patients respond normally to verbal commands. Although cognitive function and physical coordination may be impaired, airway reflexes, and ventilatory and cardiovascular functions are unaffected. Patients whose PASS Score (see Appendix 1 the Procedure and Anesthesia Scoring System) is "2" for consciousness are presumed to be minimally sedated. This policy does not establish requirements for minimal sedation / anxiolysis.

4.1.2 **Moderate Sedation/Analgesia (Moderate Sedation)** - Moderate Sedation is a drug-induced depression of consciousness during which patients respond purposefully to verbal commands, either alone or accompanied by light tactile stimulation. No interventions are required to maintain a patent airway, and spontaneous ventilation is adequate. Cardiovascular function is usually maintained. A patient who's PASS Score is "1" for

consciousness is presumed to be moderately sedated.

4.1.3 **Deep Sedation/Analgesia-** Deep Sedation is a drug-induced depression of consciousness during which patients cannot be easily aroused, but respond purposefully following repeated or painful stimulation. The ability to independently maintain ventilatory function may be impaired. Patients may require assistance in maintaining a patent airway, and spontaneous ventilation may be inadequate. Cardiovascular function is usually maintained. A patient who's PASS Score is "0" for consciousness is presumed to be deeply sedated. Deep sedation shall be intentionally induced by the Emergency Medicine practitioner with current deep sedation privileges or by others specifically approved by the Department of Anesthesiology.

4.1.4 **Anesthesia-**General anesthesia is a drug-induced loss of consciousness during which patients are not arousable, even by painful stimulation. The ability to independently maintain ventilatory function is often impaired. Patients often require assistance in maintaining a patent airway, and positive pressure ventilation may be required because of depressed spontaneous ventilation or drug-induced depression of neuromuscular function. Cardiovascular function may be impaired. General anesthesia shall be intentionally induced only by a privileged Anesthesia practitioner.

5.0 Provisions/Procedures

5.1 Privileges

5.1.1 Moderate Sedation

5.1.1.1 Practitioners who order medication intended to induce moderate sedation shall have a current moderate sedation privilege. These practitioners shall be qualified to rescue patients who enter a state of Deep sedation/Analgesia and have expertise in airway management and advanced life support. They shall have current age appropriate certification in American Heart Association Advanced Cardiac Life Support (ACLS), Pediatric Advanced Life Support (PALS), or Neonatal Resuscitation Program (NRP).

[**NOTE:** As used in this policy, "practitioner" means physician (other than anesthesiologists), dentist, podiatrist, physician assistants, certified nurse midwife or nurse practitioner. Physician assistants may only administer agents intended to induce moderate, when the supervising physician is personally present within the suite where the procedure is performed.]

5.1.2 Deep Sedation

5.1.2.1 Practitioners who order or manage patients where deep sedating agents are administered, ie, propofol, ketamine, and etomidate, shall be privileged for deep sedation. These practitioners shall be qualified to rescue patients who enter a state of General Anesthesia and be able to intubate the patient and have expertise in airway management and advanced life support. They shall have current age appropriate certification in American Heart Association Advanced Cardiac Life Support (ACLS), Pediatric Advanced Life Support (PALS), or Neonatal Resuscitation Program (NRP).

5.2 **Current Competencies**

5.2.1 Practitioner privileges for moderate and deep sedation shall be based on documentation of current clinical competence, including the successful completion of a sedation test approved by the Executive Committee.

5.2.2 RN Competencies for moderate and deep sedation shall be demonstrated initially and annually with documentation in the employee file.

5.2.3 RN Competencies includes current ACLS and PALS.

5.3 **Deep Sedation in the Emergency Department**

5.3.1 Deep Sedation in the Emergency Department shall be induced and managed according to this policy.

5.3.1.1 At least one Registered Nurse with current sedation competencies shall be present to monitor patients undergoing Deep Sedation.

5.3.1.2 Practitioners inducing deep sedation shall either be Certified by the American Board of Emergency Medicine or Certified in current age appropriate certification in American Heart Association Advanced Cardiac Life Support (ACLS), Pediatric Advanced Life Support (PALS), or Neonatal Resuscitation Program (NRP)

5.3.1.3 Deep sedation shall be intentionally induced under the direction of an Emergency Medicine practitioner with current deep sedation privileges or by others specifically approved by the Department of Anesthesiology.

5.4 **Exclusions**

5.4.1 Due to the setting and the nature of care provided, the following situations are assumed to meet the standards of care prescribed in this policy:

5.4.1.1 **Patients on a Ventilator in a Critical Care Unit**. (Rationale: This policy is intended to assure appropriate monitoring and resuscitative readiness for patients whose airway is at risk. Patients on a mechanical ventilator in a critical care unit have this need met and sedation / paralysis of these patients is often necessary to maintain mechanical ventilation. In addition, the standard of care and documentation common to critical care settings across the Region are equivalent to the standards established by this policy.)

5.4.1.2 **Infants in a Special Care Nursery**. (Rationale: The standard of care and documentation for all low birth weight and very low birth weight infants in special care nurseries is deemed to be equivalent to the monitoring and other safety precautions specified in this policy. This exclusion does not apply to infants receiving continuing or normal newborn care in a special care nursery.)

5.5 **Setting**

5.5.1 Reversal Agents

5.5.1.1 Reversal drugs shall be immediately available (including naloxone & flumazenil). Note: There is no reversal agent for propofol, ketamine or etomidate.

5.6 **Emergency Medications, Equipment and Supplies**

5.6.1 The following resuscitative equipment shall be immediately available in any location where care is provided to moderately or deeply sedated patients. Such equipment shall be regularly tested to ensure proper functioning:

5.6.1.1 Crash cart with defibrillator and advanced cardiovascular resuscitation medications and supplies approved by the appropriate Professional Staff committee,

5.6.1.2 Self-inflating resuscitation bag(s) and mask(s)

5.6.1.3 Oxygen supply with masks and nasal cannulae,

5.6.1.4 Suction set up

5.6.1.5 Cardiac monitor,

5.6.1.6 Continuous pulse oximeter,

5.6.1.7 Continuous end tidal Capnography (ETCO2 monitor)

5.6.1.8 Blood pressure device (automatic or manual)

5.6.1.9 Sedating and analgesic drugs (specified by the physician) and reversal agent readily available

5.7 **Pre-Procedure: Pre-Sedation Assessment**

5.7.1 A History and Physical (H&P) examination is required and documented prior to a procedure requiring deep sedation, unless deemed an emergency and noted by the physician. In the Emergency Department the ED note will suffice for a H&P. Allergies, medications and pertinent medical history are reviewed and documented prior to moderate sedation.

5.7.2 Informed consent is obtained from the patient by the physician and documented.

5.7.3 The nurse caring for the patient verifies and documents the following in the pre-sedation nursing assessment:

5.7.3.1 NPO Status

5.7.3.2 Physician documentation of informed consent and the patient or legal guardian has signed the hospital consent form for the procedure.

5.7.3.3 Baseline PASS Score that includes vital signs

5.7.3.4 Patient has a responsible adult to accompany them home.

5.7.3.5 Provide patient/family education related to sedation and the procedure.

5.7.3.6 Establish age appropriate peripheral IV access as ordered.

5.7.3.7 Conduct a pre-procedure verification process that verifies: the correct procedure, correct patient and the correct site.

5.7.3.8 Site marking is done by the physician if applicable.

5.7.3.9 Time out is conducted immediately prior to the procedure.

5.7.4 Pre-Sedation Assessment by the physician includes documentation of:

5.7.4.1 Intended level of sedation

5.7.4.2 ASA Score – (consider Anesthesia consult for ASA scores III, IV or E)

5.7.4.3 Heart Exam

5.7.4.4 Lung Exam

5.7.4.5 Mallampati Airway Assessment

This may include the following:

5.7.4.5.1 Inability to open mouth

5.7.4.5.2 Receding chin (micrognathia)

5.7.4.5.3 Prominent incisors

5.7.4.5.4 Morbid obesity

5.7.4.5.5 Sleep apnea

5.7.4.5.6 Poor cervical spine mobility

5.7.4.5.7 Large Tongue (macroglossia)

5.7.4.5.8 Short, muscular neck

5.7.4.5.9 Snoring

5.7.4.6 NPO Guidelines

Age	Solids & Non-Clear Fluids	Clear Fluids
0-6 months	4 hours	2 hours
6 months – 3 years	6 hours	2 hours
3 years +	6-8 hours	2 hours

5.8.5 **Exception for NPO:** When a physician assessment of risks outweighs delaying an Emergent/Urgent procedure, the physician may make the decision to proceed with appropriate documentation.

5.8 **Immediately Prior to Induction**

5.8.1 The responsible practitioner reevaluates the patient immediately before administering moderate or deep sedation.

5.8.2 The following shall be recorded as a baseline prior to intravenous administration of the moderate / deep sedating agent:

5.8.2.1 Blood pressure

5.8.2.2 Heart rate

5.8.2.3 Respirations

5.8.2.4 Oxygen saturation

5.8.2.5 End-Tidal CO2

5.8.2.6 Temperature

5.8.2.7 Procedure and Anesthesia Scoring System (PASS)

5.9 **Intra-Procedure**

5.9.1 Moderately and Deeply Sedated patients shall be subject to ongoing observation by a person with demonstrated competence in the identification of respiratory failure and other potential complications. Staff responsible for monitoring the sedated patient should have no other duties that would cause the patient to be unattended or compromise continuous monitoring.

5.9.2 Vital signs and "Level of Sedation" are monitored continuously and recorded every 5 minutes from the time of medication administration and during the procedure. These vital signs include the following:

5.9.2.1 Blood pressure

5.9.2.2 Heart rate (via cardiac monitoring)

5.9.2.3 Respirations

5.9.2.4 Oxygen Saturation

5.9.2.5 End-Tidal CO2

5.9.2.6 Level of consciousness, adequacy of ventilation; and ability to maintain a patent airway with intact protective reflexes.

5.10 **Post-Procedure**

5.10.1 Patients who receive moderate and deep sedation are monitored and the PASS score is recorded every 15mins post-procedure for 30 minutes (a longer observation is required if a reversal agent was used). The patient may be discharged after this monitoring period when these discharge criteria are met:

5.10.1.1 A PASS Score of 13 or greater, or return to baseline; and

5.10.1.2 Vital Signs are stable.

5.10.1.3 The responsible physician conducts a discharge assessment for patients who received reversal agents

5.10.2 The practitioner shall document a procedure report in the medical record immediately upon completion of the procedure and before the patient is transferred to the next level of care.

5.10.3 The physician who administered the deep sedation or his/her designee that has deep sedation privileges shall complete and document a post-sedation evaluation within 48hrs or prior to

discharge if sooner:

5.10.4 Elements of a post-sedation evaluation are:

5.10.4.1 Respiratory function, including respiratory rate, airway patency, and oxygen saturation

5.10.4.2 Cardiovascular function, including pulse and blood pressure

5.10.4.3 Mental status

5.10.4.4 Temperature

5.10.4.5 Pain

5.10.4.6 Nausea, vomiting and hydration status

5.10.5 The practitioner responsible for the sedation shall consider consultation from an anesthesiologist if a patient fails to recover from procedural sedation as anticipated.

5.10.6 Patients shall be given discharge instructions related to sedation and the procedure performed. The registered nurse or responsible practitioner shall verify that the patients possess and is aware of an emergency contact number that can be reached twenty-four hours a day, seven days a week.

5.10.7 Patients shall be discharged home in the company of a responsible adult, unless otherwise approved by the responsible practitioner.

5.11 **Quality Assessment Performance Improvement, (QAPI)**

5.11.1 Any of the following events will be referred to the Quality Improvement Committee for evaluation:

5.11.1.1 Wrong procedure, patient, side or site

5.11.1.2 Unplanned admission/transfer to higher level of care

5.11.1.3 Cardiac arrest

5.11.1.4 Use of reversal agents

5.11.1.5 Use of assistance with ventilation requiring bag-valve-mask ventilation or laryngeal or endotracheal airways

5.11.1.6 Anesthesia called for assistance

6.0 References/Appendices

6.1 Appendix 1
  6.1.1 Procedure & Anesthesia Scoring System (PASS) Chart
6.2 Appendix 2
  6.2.1 Adult Drug Dosage Guidelines for Moderate and Deep Sedation
6.3 Appendix 3
  6.3.1 Pediatric Drug Dosage Guidelines for Moderate and Deep Sedation
6.4 Advisory on Granting Privileges for Deep Sedation to Non- Anesthesiologist Sedation Practitioners (Approved by the ASA House of Delegates on October 20, 2010)
6.5 Continuum of Depth of Sedation: Definition of General Anesthesia and Levels of Sedation/Analgesia (Approved by the ASA House of Delegates on October 13, 1999, and last amended on October 15, 2014)
6.6 Statement on Safe use of Propofol (Approved by the ASA House of Delegates on October 27, 2004, and amended on October 15, 2014)
6.7 CMS §482.52 Condition of Participation: Anesthesia Services
6.8 Title 42 Code of Federal Regulations (CFR) Sections 416.42 and 482.52
6.9 ACEP policy, http://www.acep.org/workarea/DownloadAsset.aspx?id=93816
6.10 The Joint Commission Comprehensive Accreditation Manual for Hospitals 2015: Standards PC.01.02.03, PC.03.01.01, PC.03.01.03, PC.03.01.05, PC.03.01.07, RC.02.01.03

7.0 Signature Lines

Include the signature(s) of the senior regional leader(s) that approved the policy and procedure in accordance with the SCAL regional policy and procedures guidelines.

Signature: Nirav Shah **Date:** 11/6/2015
Nirav Shah, M.D.; SVP COO Clinical Operations

Appendix 1

Procedure & Anesthesia Scoring System (PASS)		
Recovery/Level of Sedation		**SCORE**
Consciousness	Awake and alert, turns toward voice	2
	Arousable but drifts back to sleep	1
	Unresponsive (except to painful/repeated stimulus)	0
Activity	Appropriate for age or development	2
	Weak for age or development	1
	No voluntary movement	0
Circulation	Stable BP within 15% of pre-sedation level	2
	BP within 30% of pre-sedation level	1
	BP>30% higher or lower than pre-sedation level	0
Respiration	Able to cough, breathe deeply or cry	2
	Dyspnea or limited breathing	1
	Apnea or obstructed breathing, requires assistance to maintain airway	0
O2 Saturation	Room Air: O2 Sat greater than 95%	2
	Needs supplemental O2 to maintain O2 Sat greater than 95%	1
	O2 Sat less than 95% with O2 supplementation	0
Readiness for Discharge		
Pain	None or mild pain	2
	Moderate or severe pain controlled with IV analgesics	1
	Persistent severe pain	0
Emetic	None or mild nausea with no vomiting	2
	Transient vomiting or retching	1
	Persistent moderate to severe nausea and vomiting	0

Appendix 2

ADULT DRUG DOSAGE GUIDELINES

- Dosages require adjustment based on patient's clinical condition

AGENT	DOSE	ACTION ONSET	ACTION DURATION
BENZODIAZEPINES			
Midazolam (Versed)	**Slow IV:** 0.5 - 1 mg (over 2 minutes) and titrate to desired effect by repeating doses every 2-3 minutes if needed **Precaution:** Reduce dose for elderly or those that have COPD or receiving concomitant narcotics. Some pt's respond to 1mg. Usual total dose: 2.5-5 mg	**IV:** 2-5 mins	Peaks at 30 – 60 minutes. Duration: 2-6 hours
Lorazepam (Ativan)	**IV:** 0.05 mg/kg, 1-4 mg IV every 10-20 mins. 4 mg max. **PO:** 1-2 mg initially. Usual dosing is 2-6 mg/day divided. May gradually increase to 10 mg daily in 2-3 divided doses **Precaution:** Monitor blood pressure and assess motor and autonomic responses	**IV:** 5-20 mins **PO:** 60 mins	**IV, PO:** 2 – 6 hours
Diazepam (Valium)	**IV:** 2.5-5 mg incremental doses of 2.5 mg can be given in 3- 4 minute intervals. Usual total 2-10 mg. **PO:** 2-10 mg 2-4 times/day **Precaution:** incompatible with most medications. Potential complications: hypotension, confusion, drowsiness & apnea	**IV:** 2-5 mins **PO:** 30 mins	Peaks at 60-90 minutes Elimination half-life 36hours
NARCOTICS			
Morphine	**IV:** 1-5 mg every 2-15 minutes. 2-5 mg IV every 5-15 mins **Precaution:** Itching & hypotension may occur	**IV:** 5-10 mins/ 15-60 mins	**IV:** 2 – 4 hours
Fentanyl (Sublimaze)	**IV:** 1 - 4 mcg/kg. Typical dose is 25-50 mcg; may repeat every 5-15 minutes. Usual total 50-200mcg **Precaution:** 100 times more potent than morphine. Rapid administration causes skeletal muscle & chest wall rigidity	**IV:** 30-60 sec	Peaks in 5-15 mins **IV:** 30-60 minutes
Meperidine (Demerol)	**IV:** 12.5-50 mg every 15 minutes. Usual dose is 50-100mg.	**IV:** 5-10 mins	**IV:** 2-4 hours
OTHER AGENT			
Diprivan (Propofol)* Injectable Emulsion for adults & children ≥2years. Adhere to strict aseptic technique during handling. A soy based product containing egg lecithin with no preservatives, can support growth of microorganisms	**IV:** 0.5-1.5mg/kg. May repeat 0.5mg/kg boluses every 3-5 mins as needed for continued sedation **Precaution:** rapid bolus injection can result in undesirable cardiorespiratory depression (apnea and hypotension). Discard unused portions at the end of the procedure or at 6 hours. Flush IV every 6 hours & at the end of the procedure to remove residual from the line	**IV:** 1-2 mins	**IV:** 3-10 minutes
REVERSAL AGENTS			
Naloxone (Narcan) Narcotic antagonist	**IV:** 0.2-0.4 mg every 2-3 minutes as needed **Precaution:** rapid reversal may cause nausea, hypertension	**IV:** 1-2 mins	**IV:** 30-60 mins short half-life 30-81 mins; may require repeat in 1-2 hours
Flumanzenil (Romazicon) Benzodiazepine antagonist	**IV:** 0.2 mg every minute up to max of 1 mg. Most patients respond to 0.6-1 mg; up to 3 mg has been reported **Precaution:** may induce seizures in pt's with seizure history	**IV:** 1-3 mins	**IV:** 30-60 mins short half-life 41-79 mins; may repeat 20 min intervals

Adapted from: ***Southern CA Regional Drug Information Services***

*These agents are restricted to practitioners with deep sedation privileges operating under guidelines approved by the Medical Executive Committee. IV = intravenous; IM = intramuscular; IN = intranasal; PO = by mouth; PR = by rectum; and SC = subcutaneous

Appendix 3

PEDIATRIC DOSAGE GUIDEINES: [1]

- Guidelines do not apply to neonates or ex-premature infants up to 6 months
- Dosages require adjustment based on patient's clinical condition
- Consider adult dosing guidelines for patient's greater than 50 kg

AGENT	DOSE	ACTION ONSET	ACTION DURATION
BENZODIAZEPINES			
Midazolam (Versed)	IV: 0.05 mg/kg, maximum of 5 mg titrated over one hour IM: 0.1 mg/kg IN: 0.2-0.4 mg/kg, maximum dose 7.5 mg PO: 0.25-0.5 mg/kg; maximum total dose 12 mg Precaution: Three times more potent than diazepam. Used with opiates can cause overdosage and complications	**IV:** 1-2 mins **IM:** 5-15 mins **IN, PO:** 10 mins	**IV, IM:** 30–60 minutes **IN, PO:** 1–2 hrs
Lorazepam (Ativan)	IM, IV: 0.05-0.1 mg/kg, maximum single dose 4 mg PO: 0.05-0.1 mg/kg, maximum single dose 2 mg Precaution: Monitor blood pressure and assess motor responses PO: 60 mins and autonomic responses	**IV:** 3-5 mins **IM:** 10-20 mins	**IV, IM, PO:** 2–6 hrs
BARBITURATES			
Pentobarbital	IV: 1-3 mg/kg; may repeat up to 6 mg/kg IM: 2-5 mg/kg	**IV:** 1-5 mins **IM:** 5-15 mins	**IV:** 15–60 mins **IM, PO:** 2–4 hrs
(Nembutal)*	PO: 2-3 mg/kg	**PO:** 15-60 mins	
Methohexital (Brevital)*	PR: 20-30 mg/kg	**PR:** 5-15 mins	**PR:** 30–90 mins
NARCOTICS			
Morphine	IV: 0.05-0.1 mg/kg Precaution: Itching & hypotension may occur	**IV:** 5-10 mins	**IV:** 2–4 hrs
Fentanyl (Sublimaze)	IV: 1-4 mcg/kg Precaution: one hundred times more potent than morphine. Rapid administration causes skeletal muscle & chest wall rigidity	**IV:** 2-3 mins	**IV:** 20–60 mins
OTHER AGENTS			
Ketamine (Ketalar)*	IV: 0.5-2 mg/kg IM: 3-4 mg/kg Adverse Effects: Increased systemic, intracranial, & intraocular pressures; hallucinogenic emergence reactions; laryngospasm; & excessive airway secretions	**IV:** 1-2 mins **IM:** 3-10 mins	**IV, IM:** 15–60 minutes
Diprivan (Propofol) * Injectable Emulsion for adults & children ≥2years. Adhere to strict aseptic technique during handling. A soy based product containing egg lecithin with no preservatives, can support growth of microorganisms	IV: 0.5-1mg/kg. May repeat 0.5mg/kg boluses every 3-5 mins as needed for continued sedation. DO NOT use for children <2 yrs old. Precaution: rapid bolus injection can result in undesirable cardiorespiratory depression (apnea and hypotension). Discard unused portions at the end of the procedure or at 6 hours. Flush IV every 6 hours & at the end of the procedure to remove residual from the line	**IV:** 1-2 mins	**IV:** 3-10 mins
REVERSAL AGENTS			
Naloxone (Narcan) Narcotic antagonist	**IV, IM, SC:** 0.01 mg/kg; may repeat in 5min to effect **Precaution:** rapid reversal may cause nausea, hypertension	**IV:** 1-2 mins **IM:** 2-5 mins	45 mins, may be shorter than duration of opiate
Flumanzenil (Romazicaon) Benzodiazepine antagonist	**IV:** 0.01 mg/kg; Max. single dose 0.2 mg; may repeat every minute up to a maximum total dose of 1 mg **Precaution:** may induce seizures in pt's with seizure history	**IV:** 1-3 mins	45-60 mins, may be shorter than duration of the benzodiazepine

[1] Adapted from: PALS Provider Manual, AHA & AAP, 2002; and guidelines from Children's Hospital of Los Angeles & Boston Children's Hospital.
*These agents are restricted to practitioners with deep sedation privileges operating under guidelines approved by the Medical Executive Committee.
IV = intravenous; IM = intramuscular; IN = intranasal; PO = by mouth; PR = by rectum; and SC = subcutaneous